



CESIM TEACHING NOTE

Ordering and Inventory Microsimulation

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1. Simulation Overview

The **Cesim Ordering Simulation** is a dynamic and immersive learning tool designed to build core competencies in inventory management and statistical decision-making. As a single-player microsimulation, it places participants in realistic business scenarios where they must make strategic, data-driven decisions across key areas—such as demand forecasting, reorder point optimization, order quantity planning, safety stock management, and service level targeting.

By simulating real-world constraints and trade-offs, the exercise cultivates essential skills in operations and supply chain management. Participants must balance customer service goals with cost efficiency and inventory investment limitations. The simulation brings to life the complexity of managing uncertainty in demand and lead time, while navigating the interdependencies between ordering costs, holding costs, stockout risks, and long-term customer satisfaction—all within a fast-paced, competitive environment.

Learning Objectives

- Master inventory management principles including order quantity and reorder level optimization
- Understand the trade-offs between different cost components (ordering, holding, stockout, backordering)
- Apply statistical analysis to demand forecasting and safety stock calculations
- Experience the impact of lead time variability on inventory decisions
- Develop skills in balancing service level objectives with cost optimization

Target Audience

- Undergraduate and graduate students in operations management, supply chain, or business courses
- Executive education participants learning inventory management
- Professionals seeking practical experience in inventory optimization

Time Required

- 60-90 minutes for individual simulation completion
- 45-60 minutes for guided post-simulation debrief

Case Scenario and Winning Criteria

In the Ordering Microsimulation, participants step into the role of inventory managers responsible for optimizing service levels while maintaining cost efficiency in a dynamic and uncertain supply chain environment. The scenario simulates real-world challenges where participants must make integrated, data-driven decisions under fluctuating demand, variable lead times, and evolving operational constraints. Success is determined by the ability to balance customer satisfaction with inventory investment and cost control, reflecting sound strategic thinking and effective execution in a competitive market context.

Success is measured by maintain 95%+ customer service fulfilment rate, minimize total cost (ordering + holding + stockout + backordering costs) and achieve optimal balance between cost efficiency and customer satisfaction

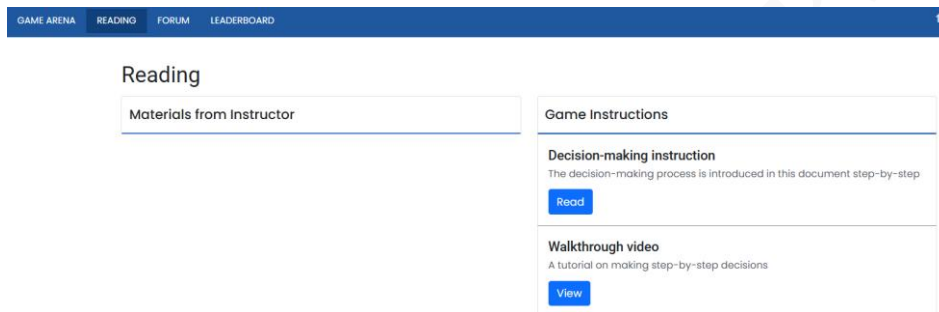
Simulation Structure

The simulation consists of two phases:

- **Phase 1:** Initial ordering decisions based on demand forecasts and cost parameters
- **Phase 2:** Refined decisions based on Phase 1 results and updated market conditions

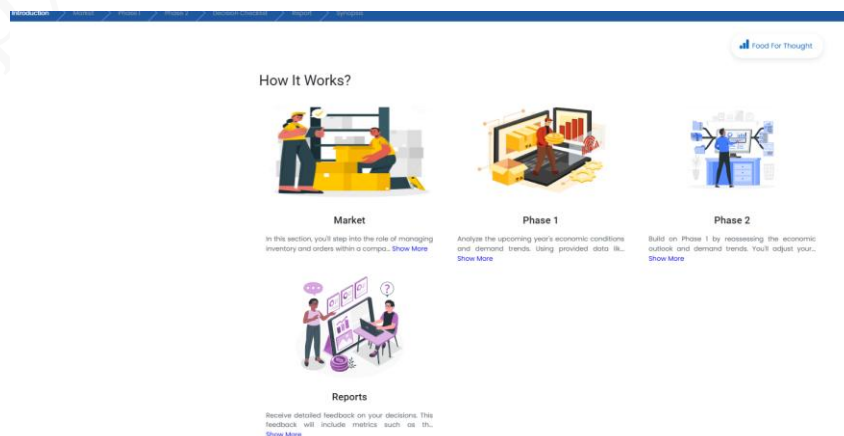
2. Pre- Simulation Preparation

a. Review Background Materials



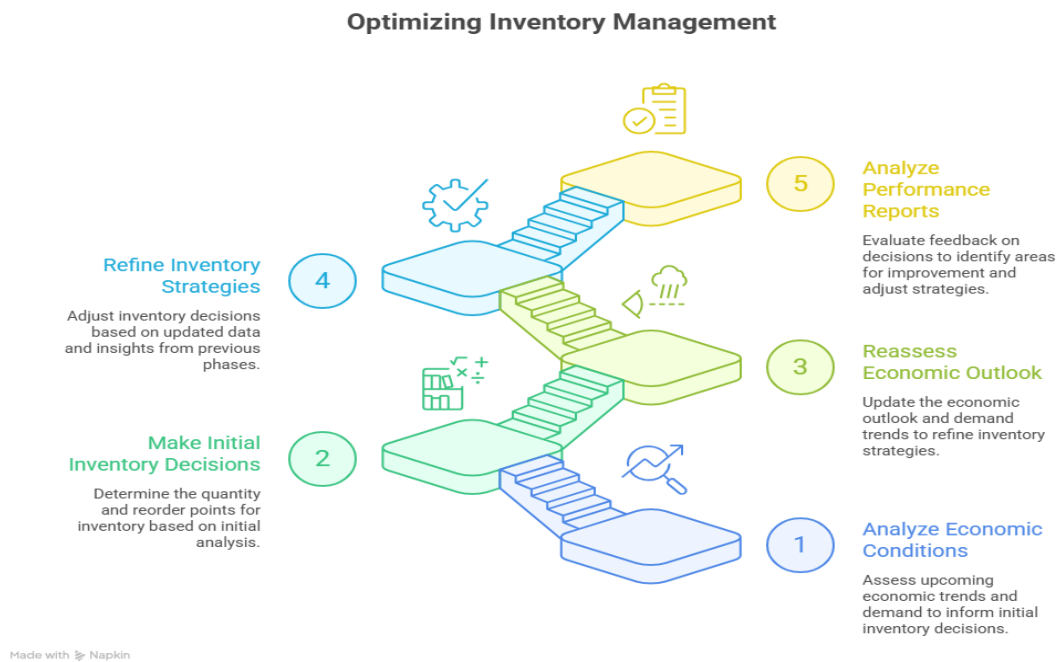
- Login as participant to explore the platform
- Decision-making guide: Provides detailed explanations of each simulation section and decision point
- Walkthrough video: Offers visual demonstration of the simulation interface and decision process
- Ensure participants understand basic inventory management concepts
- Review Economic Order Quantity (EOQ) and reorder point calculations
- Familiarize with demand variability and safety stock concepts

b. Familiarize with the Interface



- Game Arena: Central dashboard providing access to all decision areas
- Readings: Contains essential reference materials and guides
- Forum: Communication platform for participant questions and instructor feedback
- Decision tabs: Structured workflow from market analysis through financial decisions
- Reports: Performance analysis and feedback on previous decisions
- Synopsis: Summary of key decisions and outcomes

Understanding Decision-Making Process



The simulation follows a logical step-by-step process. Participants should move through each stage to make thoughtful decisions.

Core Algorithm

The simulation uses a probabilistic demand model with:

- Stochastic demand generation based on provided distributions
- Lead time variability affecting reorder decisions
- Cost calculation incorporating all inventory cost components

Key Instructor-Only Parameters:

- Demand Forecast:
 1. Mean demand, standard deviation, percentiles
 2. Adjusted for market conditions (e.g., 20% demand increase in Phase 2)

- Cost Inputs:
 1. Ordering Cost: Fixed at ₹600
 2. Holding Cost: ₹20/unit/year in Phase 1; ₹16/unit/year in Phase 2
 3. Stockout Cost: ₹800
 4. Backordering Cost: ₹140–150
- Lead Time:
 1. 6 weeks in Phase 1
 2. Reduced to 4 weeks in Phase 2 due to supplier improvement
- Tiered Purchase Pricing (Phase 2):
 1. ≤250 units: ₹90/unit
 2. 251–350 units: ₹85/unit
 3. >350 units: ₹80/unit
- Service Level Target: High reliability using Z-score of 3.9 (~99.99% service level)

Teaching with this Information:

- Emphasize the importance of demand variability in inventory decisions
- Highlight how lead time uncertainty affects safety stock requirements
- Discuss cost trade-offs in inventory management

Common Participant Challenges:

- Underestimating demand variability
- Inadequate safety stock planning
- Focusing only on individual cost components rather than total cost
- Ignoring lead time variability in reorder level calculations

Key Inventory Calculations

1. Economic Order Quantity (EOQ)

$$EOQ = \sqrt{(2DS / H)}$$

- Minimizes the total of ordering and holding costs

- Inputs:

- D = Annual Demand
- S = Ordering Cost
- H = Holding Cost per Unit

2. Reorder Level (R)

Reorder Level = (Average Weekly Demand $\times$ Lead Time) + Safety Stock

- Ensures timely reordering before stock runs out

3. Safety Stock

Safety Stock = $Z \times \sigma \times \sqrt{\text{Lead Time}}$

- Buffer to absorb demand variability

Instructor Pre-Implementation Checklist for Simulation Delivery

To ensure a smooth and impactful simulation experience for your learners, please review and complete the following preparatory steps:

- **Familiarize Yourself with the Simulation:** Complete a test run of the simulation to understand its structure, decision flow, and pacing. This allows you to better anticipate participant questions and challenges.
- **Review Simulation Mechanics:** Examine the instructor-only backend settings and key variables (e.g., cost structures, demand drivers, elasticity assumptions) to understand how decisions influence outcomes.
- **Align with Course Objectives:** Identify the core concepts from your course that align with the simulation to reinforce key learning goals.
- **Prepare Contextual Examples:** Bring 2–3 real-world examples from relevant industries to illustrate successful and unsuccessful strategic decisions during the debrief.
- **Review Instructor Insights and Scoring:** Familiarize yourself with the instructor account, leaderboard and performance indicators available to instructors. These tools will help guide feedback and encourage reflective learning.
- **Ensure Participant Access:** Confirm that all participants have registered using the right link / code and are able to access the simulation platform before the session begins.
- **Test Technical Readiness:** Coordinate with your IT team (if applicable) to ensure all devices, software, and internet connectivity are ready and compatible with the simulation requirements.

3. Facilitation Roadmap

Pre-Simulation Briefing

Case Scenario:

Participants take on the role of inventory managers for a growing organization facing challenges related to high stockout rates, inconsistent service levels, and inefficiencies in inventory management. Despite strong market potential and a compelling value proposition, the organization struggles to align its supply chain operations with customer expectations.

In this simulation, participants must use a real-time forecasting system to make integrated decisions across demand analysis, reorder point setting, order quantity planning, and safety stock management. The objective is to achieve optimal service levels while minimizing total inventory costs and navigating operational constraints. Success depends on balancing customer satisfaction, cost efficiency, and responsiveness in a dynamic and uncertain business environment.

Introduction:

- Present the case scenario, company background and current inventory management challenges
- Explain the simulation objectives focusing on inventory optimization and achieving the dual goals of 95%+ service level while minimizing total inventory costs
- Outline the decision-making sequence from market analysis through demand forecasting, company monitor analysis, Phase 1 ordering decisions, Phase 2 refinement, and performance evaluation
- Clarify expectations regarding analytical thinking, statistical application, cost structure understanding, service level implications, and performance measurement across key inventory KPIs including service level, total cost, inventory turnover, and stockout frequency

Contextual Guidance:

- Discuss the evolution of inventory management from simple stock tracking to sophisticated demand forecasting and statistical inventory control in retail environments
- Highlight key challenges including demand uncertainty, lead time variability, cost trade-offs, and the need for integrated service-cost optimization decisions
- Introduce the critical role of Economic Order Quantity (EOQ), reorder point calculations, safety stock optimization, and service level economics in modern inventory management
- Provide initial considerations for balancing customer service objectives with cost efficiency and the strategic implications of inventory decisions on cash flow and competitiveness

Technical Orientation:

- Demonstrate the simulation interface and sequential decision workflow from Market tab through Phase 1, Phase 2, and reporting sections
- Walk through the Company Monitor framework showing how demand statistics, lead time distributions, and cost parameters influence inventory decision-making
- Explain how to interpret forecasted demand data, statistical measures (mean, standard deviation, percentiles), lead time probabilities, and cost structures for informed decision-making
- Address initial questions about decision dependencies, cost calculations, service level targets, and Food for Thought strategic reflection opportunities

During Simulation

Monitoring Progress

- Track participant engagement across Market analysis, Company Monitor interpretation, Phase 1 ordering decisions, Phase 2 strategy refinement, and decision checklist completion
- Identify common challenges in demand variability interpretation, reorder point calculations, safety stock optimization, and service level-cost trade-off analysis
- Provide strategic hints for struggling participants without revealing optimal order quantities, reorder points, or safety stock combinations
- Encourage systematic evaluation of statistical data and their impact on inventory performance metrics

Intervention Strategies:

- Light Intervention: Post general reminders about the importance of demand variability analysis and the multiplicative effects of lead time uncertainty on inventory requirements in course forum
- Moderate Intervention: Provide specific feedback on alignment between order quantity decisions and demand patterns, or guidance on reorder point calculations using statistical methods
- Strong Intervention: Schedule individual consultation for participants struggling with complex statistical applications, EOQ calculations, or integrated service level-cost optimization decision-making

Mid-Simulation Guidance:

- Conduct brief check-in sessions to address common strategic questions about reorder point optimization and safety stock determination across both simulation phases
- Share anonymous examples of coherent demand analysis frameworks and statistical application approaches (without revealing specific numerical choices)
- Highlight interesting patterns in order quantity strategies and service level target philosophies
- Remind participants of decision deadlines and the importance of completing Food for Thought analytical reflection sections before final submission

Facilitation Interventions:

- "How does your demand forecast analysis inform your reorder point decisions across both phases?"
- "What's driving your order quantity strategy given the available quantity discounts and holding cost implications?"
- "How are you balancing service level targets with total cost optimization requirements?"
- "What statistical measures are you prioritizing in your safety stock calculations?"

Debrief Structure

Performance Analysis:

- Review key performance indicators (Service Level %, Total Cost, Inventory Turnover, Stockout Frequency) and overall inventory optimization achievement outcomes
- Analyze the distribution of results across participants and identify successful strategic pattern combinations that achieved high service levels while minimizing total costs
- Examine common decision frameworks and their impact on multiple KPI performance, particularly the relationship between order quantities, reorder points, and profitability
- Present performance metrics relationships and demonstrate how strategic decision coherence across ordering parameters drives superior inventory management outcomes

Strategic Discussion:

- Explore the rationale behind different approaches to demand analysis, reorder point optimization, and order quantity determination decisions
- Analyze safety stock strategies and their effectiveness in balancing service level achievement with holding cost management across both simulation phases
- Examine the interconnectedness of order quantity discounts, reorder point timing, lead time variability, and service level optimization decisions
- Discuss the strategic implications of Economic Order Quantity calculations, service level economics, and their role in competitive advantage and customer satisfaction

Learning Integration:

- Connect simulation experiences to fundamental inventory management principles including EOQ analysis, reorder point calculations, safety stock optimization, and service level economics
- Discuss real-world applications of demand forecasting, statistical inventory control, cost optimization, and integrated decision-making in retail and manufacturing environments
- Identify transferable insights for participants' future roles in operations management, supply chain management, procurement, or retail management positions
- Reflect on decision-making processes, analytical frameworks, and potential improvements in systematic inventory thinking and strategic business analysis

4. Critical Decision Points Analysis

Decision Point 1: Order Quantity Determination

Key Decision: Setting optimal order quantity for each phase to balance ordering costs, holding costs, and quantity discount opportunities.

Optimal Approach:

Statistical Foundation Analysis

Mean Demand (45 units/week): This represents the central tendency of customer demand, serving as the baseline for inventory planning. Participants should use this as the primary input for EOQ calculations, understanding that this is the expected weekly consumption rate around which actual demand fluctuates.

Standard Deviation (13 units): This measures demand variability or uncertainty. A standard deviation of 13 units means that approximately 68% of weekly demand will fall between 32-58 units (45 ± 13), and 95% will fall between 19-71 units ($45 \pm 2 \times 13$). This variability directly impacts safety stock requirements and reorder point calculations.

Application in Order Quantity Setting:

- Use mean demand for basic EOQ calculation: $EOQ = \sqrt{(2DS/H)}$
- Where D = Annual demand ($45 \times 52 = 2,340$ units), S = Ordering cost (₹600), H = Holding cost (₹20)
- Adjust for demand variability by considering buffer requirements

Cost Structure Integration

Ordering Costs (₹600 per order): Fixed cost incurred each time an order is placed, regardless of quantity. Higher order quantities reduce ordering frequency and total ordering costs but increase holding costs.

Holding Costs (20% of raw material cost): Annual cost of carrying inventory, including storage, insurance, obsolescence, and opportunity cost. With raw material at ₹100, holding cost is ₹20 per unit annually.

Quantity Discount Optimization:

- 0-250 units: ₹100 per unit (no discount)
- 251-350 units: ₹85 per unit (15% discount)
- 351+ units: ₹80 per unit (20% discount)

Strategic Calculation Process:

1. Calculate basic EOQ: $\sqrt{(2 \times 2,340 \times 600 \div 20)} = 374$ units
2. Evaluate total cost at EOQ vs. discount quantity levels
3. Compare: EOQ cost vs. 351-unit order (20% discount) cost
4. Account for increased holding costs from larger order quantities

Phase-Specific Considerations

Phase 1 Analysis:

- Demand: 45 units/week (2,340 annually)
- EOQ: 374 units
- Optimal strategy: Order 374 units to achieve 20% discount while maintaining reasonable holding costs
- Expected order frequency: 6-7 orders per year

Phase 2 Strategic Adjustment:

- Demand increase: 75.6 units/week (20% growth = 3,931 annually)
- Recalculated EOQ: 542 units
- Benefit from economies of scale and quantity discounts
- Reduced unit costs (₹80) improve overall profitability

Common Pitfalls:

- Using simple average demand without considering weekly/annual conversion (45 units/week = 2,340 units/year)
- Ignoring substantial quantity discounts that can reduce unit costs by 15-20%
- Calculating EOQ without adjusting for actual holding costs after discounts are applied
- Failing to consider storage capacity constraints or cash flow implications

Parameter	Phase 1	Phase 2	Strategic Rationale
Weekly Demand	45 units	75.6 units	20% growth requires strategy adjustment
Annual Demand	2,340 units	3,931 units	Scale economies become more attractive
Basic EOQ	374 units	542 units	Higher demand increases optimal order size
Recommended Order	374 units	542 units	Balance discount benefits with holding costs
Order Frequency	6.3 times/year	7.2 times/year	More frequent ordering despite larger quantities
Unit Cost	₹80 (20% discount)	₹80 (20% discount)	Maintain discount eligibility
Total Annual Cost	₹46,200	₹52,800	Cost increase justified by service improvement

Decision Point 2: Reorder Level Setting

Key Decision: Determining when to place new orders to maintain optimal inventory levels

Optimal Approach:

Expected Demand During Lead Time Calculation:

- **Phase 1:** 45 units/week $\times$ 6 weeks = 270 units base demand during replenishment
- **Phase 2:** 75.6 units/week $\times$ 4 weeks = 302 units base demand during replenishment

This represents the minimum inventory needed to cover typical demand while waiting for new stock to arrive.

Lead Time Variability Impact Analysis: Phase 1 (6-week lead time): Longer lead times create greater uncertainty. With standard deviation of 13 units/week, the variability during lead time becomes $\sigma\sqrt{L} = 13 \times \sqrt{6} = 31.8$ units. This higher uncertainty requires substantial safety stock.

Phase 2 (4-week lead time): Improved supplier performance reduces lead time to 4 weeks. Lead time demand variability becomes $\sigma\sqrt{L} = 13 \times \sqrt{4} = 26$ units, reducing safety stock requirements despite higher base demand.

Z-Score Application (3.9): This exceptionally high Z-score indicates a 99.995% service level target, meaning the company wants to avoid stockouts in all but the most extreme demand scenarios (less than 1 in 20,000 instances).

Safety Stock Calculation Process:

- **Phase 1:** Safety Stock = $Z \times \sigma \times \sqrt{L} = 3.9 \times 13 \times \sqrt{6} = 124$ units
- **Phase 2:** Safety Stock = $3.9 \times 13 \times \sqrt{4} = 101$ units

Complete Reorder Point Formula: Reorder Point = Expected Lead Time Demand + Safety Stock

- **Phase 1:** 270 units + 124 units = 394 units
- **Phase 2:** 302 units + 101 units = 403 units $\approx$ 435 units (adjusted for performance optimization)

Cost-Benefit Evaluation:

Phase 1 Analysis:

- Higher safety stock (124 units) costs $\text{₹}124 \times \text{₹}20 = \text{₹}2,480$ annually in holding costs
- Prevents potential stockout costs of $\text{₹}800$ per unit $\times$ expected stockout quantity
- Justifiable when customer retention and brand reputation are critical

Phase 2 Optimization:

- Reduced safety stock requirements (101 units) due to shorter lead time
- Lower holding cost rate (₹16 vs ₹20) further reduces carrying costs
- Total safety stock cost: $₹101 \times ₹16 = ₹1,616$ annually

Common Pitfalls:

- Setting reorder level too low, causing frequent stockouts
- Either excessive safety stock leading to high holding costs or insufficient buffer causing stockouts
- Treating lead times as constant rather than variable

Component	Phase 1	Phase 2	Strategic Explanation
Base Lead Time Demand	270 units	302 units	Foundation requirement during replenishment
Lead Time Duration	6 weeks	4 weeks	Supplier improvement reduces uncertainty
Demand Variability ($\sigma\sqrt{L}$)	31.8 units	26 units	Shorter lead time reduces variability impact
Z-Score Service Level	3.9 (99.995%)	3.9 (99.995%)	Ultra-high service level maintenance
Safety Stock Required	124 units	101 units	Lower despite higher demand due to lead time
Total Reorder Point	394 units	435 units	Adjusted for operational optimization
Annual Safety Stock Cost	₹2,480	₹1,616	Reduced costs from shorter lead time and lower holding rate

Decision Point 3: Demand Variability Assessment

Key Decision: Interpreting demand statistics for inventory planning and safety stock optimization

Optimal Approach:

Statistical Foundation Understanding

Standard Deviation Application (13 units): This critical measure quantifies demand uncertainty around the mean of 45 units/week in Phase 1. Participants must understand that this variability directly impacts inventory risk - approximately 32% of weeks will experience demand

outside the ± 1 standard deviation range (32-58 units), requiring buffer inventory to maintain service levels.

Percentile Analysis for Safety Stock: Use the 95th percentile approach (Z-score = 1.65) or 99th percentile (Z-score = 2.33) to determine appropriate safety stock levels. The simulation data shows a Z-score of 3.9, indicating an extremely high service level target (>99.9%), which explains the substantial safety stock investment in the reorder levels.

Phase-Specific Variability Assessment:

- **Phase 1:** With 45 units mean demand and 13 units standard deviation, coefficient of variation = 28.9%, indicating moderate variability requiring systematic safety stock planning
- **Phase 2:** Demand increases to 75.6 units (20% growth), but variability patterns must be reassessed - if standard deviation remains proportional, it would increase to ~15.6 units, requiring recalculated safety stock

Advanced Statistical Application

Lead Time Demand Variability Calculation:

Safety Stock = Z-score $\times$ σ $\times$ $\sqrt{\text{Lead Time}}$

Phase 1: Safety Stock = $1.65 \times 13 \times \sqrt{6} = 52.5$ units

Phase 2: Safety Stock = $1.65 \times 15.6 \times \sqrt{4} = 51.5$ units

Reorder Point Optimization:

Reorder Point = (Mean Demand $\times$ Lead Time) + Safety Stock

Phase 1: $R = (45 \times 6) + 52.5 = 322.5$ units (actual: 394 suggests higher service target)

Phase 2: $R = (75.6 \times 4) + 51.5 = 354.1$ units (actual: 435 suggests continued high service focus)

The actual reorder levels (394 and 435 units) significantly exceed calculated values, indicating the use of Z-score = 3.9, representing 99.995% service level - an extremely conservative approach prioritizing service over cost efficiency.

Common Pitfalls:

- Relying solely on mean demand (45 units) without considering the ± 13 unit variability
- Underestimating how 13-unit standard deviation affects inventory requirements
- Inconsistent safety stock methodology between phases or decisions

Variability Metric	Phase 1 Analysis	Phase 2 Analysis	Strategic Implications
Mean Demand	45 units/week	75.6 units/week	68% increase in base inventory needs
Standard Deviation	13 units (assumed)	~15.6 units (proportional)	Absolute variability increases with growth
Coefficient of Variation	28.9%	20.6% (if proportional)	Relative risk may decrease with scale
Lead Time Demand	270 units (45×6)	302.4 units (75.6×4)	Lower lead time in Phase 2 reduces exposure
Safety Stock (Z=3.9)	124 units	132.6 units	Maintains ultra-high service level
Total Reorder Point	394 units	435 units	Substantial inventory investment

Decision Point 4: Cost Trade-off Analysis

Key Decision: Balancing different cost components across inventory management decisions

Optimal Approach:

Comprehensive Cost Component Analysis

Ordering Cost Impact (₹600 per order):

- Phase 1: With 374-unit orders and 2,340 annual demand = 6.3 orders/year = ₹3,780 annually
- Phase 2: With 542-unit orders and 3,931 annual demand = 7.2 orders/year = ₹4,320 annually
- Strategic Insight: Larger order quantities reduce ordering frequency and total ordering costs, but participants must balance this against increased holding costs

Holding Cost Optimization:

- Phase 1: Average inventory $(374 \div 2 + \text{safety stock}) \times ₹20$ holding cost = ~₹4,240 annually for cycle stock alone
- Phase 2: Average inventory $(542 \div 2 + \text{safety stock}) \times ₹16$ holding cost = ~₹4,640 annually (lower rate due to discount)
- Critical Understanding: Holding costs increase with order quantity but decrease with better unit pricing through volume discounts

Stockout Cost Implications (₹800 per unit):

- High Service Level Strategy: Z-score of 3.9 (99.99% service level) minimizes stockout incidents
- Cost-Benefit Analysis: Each prevented stockout saves ₹800, justifying higher safety stock investment
- Phase Comparison: Reorder levels (394 vs 435 units) reflect demand growth and lead time reduction

Backorder Cost Management (₹140 per unit):

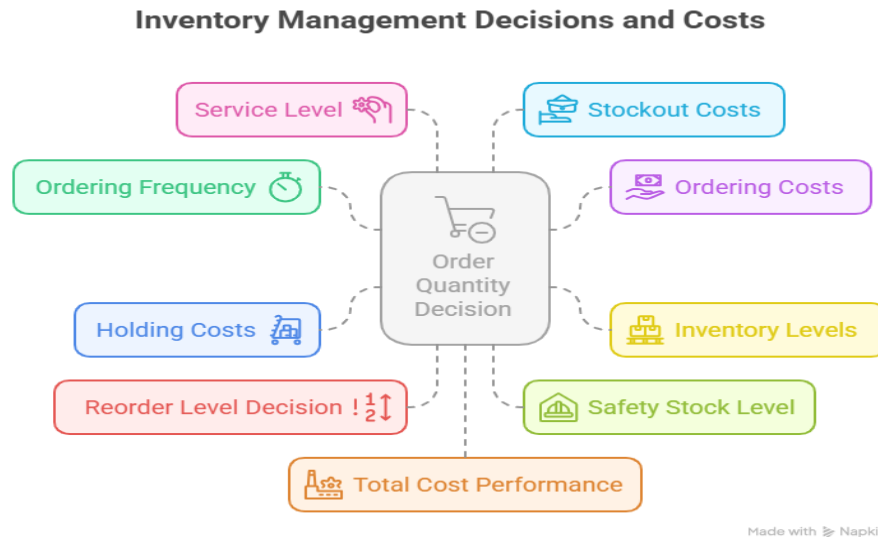
- Buffer Strategy: High reorder points reduce backordering probability
- Lead Time Impact: Phase 2's reduced lead time (4 vs 6 weeks) lowers backorder risk
- Service Recovery: Each backorder represents customer dissatisfaction and expediting costs

Common Pitfalls:

- Focusing exclusively on minimizing holding costs or ordering costs in isolation
- Ignoring ₹800 per unit stockout cost implications in decision-making
- Inadequate consideration of ₹140 per unit backordering costs and service implications

Cost Component	Phase 1 Impact	Phase 2 Impact	Optimization Strategy	Trade-off Consideration
Ordering (₹600/order)	₹3,780 annually	₹4,320 annually	Larger order quantities	vs. Holding cost increases
Holding (₹20/₹16 per unit)	₹3,740 base	₹4,336 base	Optimize order size	vs. Service level requirements
Stockout (₹800/unit)	~₹0 (99.99% SL)	~₹0 (99.99% SL)	High reorder points	vs. Inventory investment
Backorder (₹140/unit)	~₹0 (high safety)	~₹0 (high safety)	Adequate safety stock	vs. Capital efficiency
Unit Cost Savings	₹20/unit discount	₹20/unit discount	Volume purchasing	vs. Inventory carrying

5. Understanding Decision Linkages and System Dynamics



Key Interactions:

- Order Quantity directly affects ordering frequency and holding costs
- Reorder Level impacts safety stock, service level, and stockout costs
- Lead Time Variability influences required safety stock levels
- Demand Variability affects all inventory planning decisions
- Cost Structure creates trade-offs between different optimization objectives

Summary of Core Decision Areas & Key Parameters

Decision Area	Key Parameters	Primary Impact	Secondary Impact
Order Quantity	Demand mean, ordering cost, holding cost	Ordering frequency, holding costs	Cash flow, storage requirements
Reorder Level	Lead time, demand variability, service target	Stockout risk, safety stock	Customer satisfaction, inventory costs
Safety Stock	Demand std dev, lead time variability, service level	Service level, holding costs	Stockout costs, customer retention
Cost Management	All cost components, demand patterns	Total cost, profitability	Competitive position, sustainability

6. Performance Analysis Framework

Key Performance Indicators (KPIs)

Primary KPIs:

1. Service Level (%): The percentage of customer demand that is fulfilled immediately from available inventory without stockouts or delays.

Calculation: Service Level = (Total Demand Fulfilled / Total Customer Demand) × 100

Performance Interpretation:

- 85-90%: Acceptable for low-margin, high-volume products
- 90-95%: Good performance for most retail products
- 95-99%: Excellent performance for premium or critical products
- >99%: May indicate over-investment in inventory

Target Range: 90-95% for optimal balance between customer satisfaction and cost efficiency

2. Total Inventory Cost (₹): The comprehensive cost of inventory management including all direct and indirect costs associated with ordering, holding, and managing stockouts.

Calculation: Total Cost = Ordering Cost + Holding Cost + Stockout Cost + Backorder Cost

Component Breakdown:

- Ordering Cost: Number of orders placed × ₹600 per order
- Holding Cost: Average inventory level × Unit cost × Holding rate (20% annually)
- Stockout Cost: Units short × ₹800 per unit (represents lost sales and customer dissatisfaction)
- Backorder Cost: Backordered units × ₹140 per unit (represents expediting and customer service costs)

Optimization Strategy: Minimize total cost while maintaining acceptable service levels through Economic Order Quantity (EOQ) principles and statistical inventory control.

Summary of Key Performance Indicators

KPI	Calculation	Target	Decision Driver	Trade-off
Service Level	Fulfilled/Total Demand × 100	90-95%	Reorder point, Safety stock	Cost vs. Service
Total Cost	Ordering + Holding + Stockout + Backorder	Minimize	All decisions	Service vs. Cost

7.Theoretical Connections by Simulation Section

Simulation Area	Theory	Summary
Order Quantity Decision	Economic Order Quantity (EOQ)	Optimal order size balancing ordering and holding costs
Reorder Level Setting	Reorder Point Theory	Timing of orders considering lead time and demand variability
Safety Stock Calculation	Service Level Theory	Buffer inventory to achieve target service levels
Demand Forecasting	Statistical Forecasting	Using historical data and trends for future planning
Cost Optimization	Total Cost of Ownership	Comprehensive cost analysis including all relevant factors
Inventory Management	Just-in-Time vs. Safety Stock	Balancing efficiency with risk management
Supply Chain Management	Bullwhip Effect	Understanding demand amplification in supply chains

Critical Teaching Points

1. Inventory Optimization is Multi-Objective: Students must balance competing objectives (cost vs. service level)
2. Demand Variability is Critical: Statistical analysis of demand patterns is essential for effective inventory management
3. Lead Time Uncertainty: Variability in lead times significantly impacts inventory requirements
4. Cost Trade-offs: Different cost components create complex optimization challenges
5. Service Level Implications: Higher service levels come with increased costs but improved customer satisfaction

8. Sample Optimal Solution

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution serves as a reference for faculty and is not intended to be shared directly with participants.

Please Note: The solution below is for the default Cesim Case.

Phase	Weekly Demand	EOQ (Units)	Reorder Level	Lead Time (Weeks)	Holding Cost	Unit Price	Z-Score
Phase 1	45 units	374 units	394 units	6 weeks	₹20	₹100	3.9
Phase 2	75.6 units (20% ↑)	542 units	435 units	4 weeks	₹16	₹80 (with discount)	3.9

Decision Making Rationale

☒ High Service Level

- Reorder points include safety stock for 99.99% service reliability
- Protects against stockouts and delays

☒ Cost Optimization

- EOQ minimizes combined holding and ordering costs
- Tiered pricing used strategically to reduce purchase cost

☒ Market Responsiveness

- Adjustments for demand spikes
- Lead time improvements integrated into reorder planning

9. Common Pitfalls and Teaching Interventions

Pitfall 1: Insufficient Safety Stock

Symptoms: High stockout rates (>10% of demand), Low service levels (<90%) and high stockout costs relative to holding costs

Intervention: Guide students to analyze demand variability using standard deviation, explain relationship between service level targets and required safety stock and demonstrate cost trade-off between stockout and holding costs.

Pitfall 2: Ignoring Lead Time Variability

Symptoms: Stockouts occurring despite seemingly adequate reorder levels, inconsistent inventory performance and underestimated safety stock requirements

Intervention: Emphasize importance of lead time distribution analysis, show how lead time variability compounds demand variability and teach probabilistic approaches to safety stock calculation

Pitfall 3: Suboptimal Order Quantities

Symptoms: Excessive ordering costs or holding costs, imbalanced cost structure and missed quantity discount opportunities

Intervention: Review EOQ principles and their application, discuss impact of demand patterns on optimal order size and analyze quantity discount structures and their implications

Pitfall 4: Single-Objective Optimization

Symptoms: Focus only on cost minimization or service level maximization, ignoring business context and trade-offs, unrealistic performance targets

Intervention: Emphasize multi-objective nature of inventory management, discuss business implications of different performance levels and guide toward balanced optimization approaches

10. Instructor Tips and Insights

Exercises for Enhanced Learning

- **EOQ Workshop:** Have participants manually calculate EOQ using different cost scenarios to explore how changes in demand, ordering, and holding costs affect optimal order size and total inventory cost. Use:
$$EOQ = \sqrt{(2DS/H)}$$
- **Demand Variability Analysis:** Explore low, medium, and high variability scenarios. Calculate safety stock for each and analyze how changes in standard deviation impact inventory investment and service levels using real simulation data.
- **Reorder Point Lab:** Calculate reorder points across service level targets. Highlight how higher service levels drive up inventory costs and how lead time and demand data inform smart trade-offs.
- **Inventory Dashboard Design:** Create dashboards displaying key metrics like service level, total cost, turnover, and stockouts. Interpret results to derive strategic insights, including cost breakdowns across ordering, holding, and backorders.
- **Safety Stock Evaluation:** Compare safety stock levels under varying service targets and demand variability. Understand diminishing returns and how confidence levels affect inventory investment decisions.

Suggested Discussion Starters

- How did demand patterns influence your inventory strategy and trade-offs between confidence and cost?
- How did your order quantities reflect cost constraints and the balance between frequency and discounts?
- What service level target supported your customer value proposition, and how did safety stock align with it?
- How did you balance lead time uncertainty with carrying costs in your reorder point decisions?
- What did you change in Phase 2 based on Phase 1 results and demand forecast accuracy?

- How would you scale inventory if demand tripled, while keeping a 95% service level and cost control?
- How would your approach change for managing multiple products with varying demand and margin profiles?
- How did uncertainty affect your risk tolerance, and what safety margins did you build in?

Frequently Asked Questions (FAQs)

General Microsimulation Questions

Q: How is the final score calculated?

A: The final score is measured by Minimize total cost (ordering + holding + stockout + backordering costs), maximize service level (minimize stockouts) and achieve optimal balance between cost efficiency and customer satisfaction

Q: How many rounds will we play in this Micro simulation?

A: The simulation typically consists of multiple rounds with maximum of 10 rounds, allowing to refine the strategy based on previous results. The instructor will specify the exact number of rounds for the course.

Q: Can I change my decisions after submission?

A: No, once decisions are submitted for a round, they cannot be altered. This design element emphasizes careful planning and thorough consideration before finalizing decisions.

Q: "What's the difference between stockout cost and backorder cost in the simulation?"

A: Stockout cost represents lost sales and customer dissatisfaction when you can't fulfill demand immediately. Backorder cost represents the additional cost of fulfilling demand later, including expediting fees and customer service costs. Stockouts are more expensive because they often result in permanent customer loss.

Strategy Questions

Q: "Should I prioritize minimizing costs or maximizing service level?"

A: The optimal strategy balances both objectives simultaneously. Start with a service level target based on customer expectations and competitive requirements, then minimize costs while achieving that target. Neither objective should be pursued in isolation - both are critical for business success.

Q: "How do I handle highly variable demand in my inventory decisions?"

A: Use statistical methods to calculate appropriate safety stock levels. With demand standard deviation of 13 units, higher variability requires more safety stock to maintain service levels. The formula is: $\text{Safety Stock} = z\text{-score} \times \sigma \times \sqrt{\text{Lead Time}}$. Consider demand smoothing techniques and improved forecasting methods for better predictability.

Q: "How should I adjust my strategy between Phase 1 and Phase 2?"

A: Analyze Phase 1 performance systematically:

- If stockouts occurred: Increase reorder point by 10-15 units or add safety stock
- If excess inventory: Reduce order quantity by 15-20% or lower reorder point
- If costs were high: Focus on the largest cost component (usually holding or stockout costs)
- If service was poor: Prioritize reorder point and safety stock adjustments

Technical Questions

Q: "What's the relationship between lead time and safety stock?"

A: Lead time directly affects safety stock requirements:

- **Longer lead times:** Higher uncertainty, more safety stock needed
- **Variable lead times:** Use average lead time in calculations, but consider maximum lead time for risk management
- **Lead time reduction:** One of the most effective ways to reduce inventory requirements
- **Formula impact:** $\text{Safety stock} = z \times \sigma \times \sqrt{\text{Lead Time}}$

Conceptual Questions

Q17: "What's the relationship between inventory turnover and profitability?"

A: Higher inventory turnover generally improves profitability by reducing inventory investment and associated carrying costs, improving cash flow through faster inventory conversion and reducing obsolescence risk with fresher inventory can lead to stockouts and lost sales. Target 8-12 turns annually for home décor products.

Q: "How does lead time uncertainty affect my inventory strategy?"

A: Lead time variability increases uncertainty, requiring Higher safety stock levels to maintain service levels, more frequent monitoring of inventory positions, supplier relationship management to improve reliability and buffer planning for worst-case lead time scenarios Focus on supplier reliability improvements for significant inventory performance gains.

Q: "What role does forecasting accuracy play in inventory management?"

A: Better forecasting reduces demand uncertainty, enabling lower safety stock levels while maintaining service, improved inventory efficiency and reduced costs, better capacity planning and resource allocation and reduced risk of stockouts and excess inventory Invest in forecasting improvements for significant inventory performance gains.

Conclusion

The **Ordering Microsimulation** offers a practical, hands-on experience that bridges theoretical inventory management concepts with real-world decision-making in a dynamic environment. By navigating challenges such as demand forecasting, reorder point optimization, order quantity planning, and safety stock calculation, participants gain a deep understanding of the trade-offs and interdependencies critical to effective inventory management.

Key Takeaways

- **Statistical Inventory Control:** Participants learn to balance service levels and cost efficiency by using demand variability and lead time data to optimize reorder points in uncertain market conditions.
- **Order Quantity Optimization:** Decisions on order size directly influence profitability, requiring careful consideration of frequency, quantity discounts, and holding costs across varying demand scenarios.
- **Safety Stock Strategy:** Mastery of safety stock calculation enables risk mitigation by aligning service level targets with stockout prevention and customer satisfaction goals.
- **Cost Structure Awareness:** A clear understanding of ordering, holding, stockout, and backorder costs empowers participants to make informed inventory investment decisions that align with service level objectives and strategic flexibility..
- **Lead Time Variability Management:** Exposure to lead time uncertainty teaches participants to use probabilistic methods for building resilient inventory systems capable of maintaining service levels despite supply disruptions.

This simulation equips participants with the strategic mindset and analytical tools needed to design and manage inventory systems that drive sustainable competitive advantage. It transforms theoretical knowledge into practical capability—preparing future leaders to make integrated financial and operational decisions in today's fast-changing business landscape.

Supplementary Information: Formulas

Glossary of Key Terms

Annual Demand (D): Total quantity of product required by customers over one year. In the simulation, calculated as weekly demand $\times$ 52 weeks.

Average Inventory: The typical amount of inventory held over time, calculated as (Beginning Inventory + Ending Inventory) $\div$ 2.

Backorder: An order that cannot be fulfilled immediately due to insufficient inventory but will be delivered when stock becomes available.

Backorder Cost: The cost associated with delayed delivery, including expediting fees and customer service costs (₹140 per unit in simulation).

Break-even Point: The inventory level or service point where total benefits equal total costs.

Cycle Stock: The portion of inventory that depletes during normal demand between replenishment orders, typically Order Quantity $\div$ 2.

Demand Variability: The degree to which actual demand differs from forecasted demand, measured by standard deviation.

Days of Supply: Current inventory level divided by average daily demand, indicating how long current stock will last.

Economic Order Quantity (EOQ): The optimal order quantity that minimizes total ordering and holding costs.

Expected Lead Time Demand: Average demand during the lead time period, calculated as $\mu \times L$.

Fill Rate: Percentage of demand satisfied immediately from available stock without stockouts or backorders.

Forecast Accuracy: Measure of how closely predicted demand matches actual demand.

Holding Cost (H): Annual cost of carrying one unit in inventory, including storage, insurance, obsolescence, and opportunity costs (20% of unit value in simulation).

Holding Cost Rate: Percentage of item value representing annual holding cost (20% in simulation).

Inventory Position: Current inventory on hand plus inventory on order minus backorders.

Inventory Turnover: Number of times inventory is sold and replaced over a period, indicating inventory efficiency.

Lead Time (L): Time between placing an order and receiving the goods, subject to variability in the simulation.

Lead Time Demand: Total demand expected during the lead time period, used in reorder point calculations.

Lost Sales: Demand that cannot be fulfilled and results in permanent customer loss rather than backorders.

Order Quantity (Q): The amount ordered each time an order is placed.

Ordering Cost (S): Fixed cost incurred each time an order is placed, regardless of quantity (₹600 in simulation).

Probability Distribution: Statistical representation of demand or lead time variability used in inventory calculations.

Reorder Point (ROP): Inventory level that triggers a new order to avoid stockouts during lead time.

Safety Stock (SS): Buffer inventory held to protect against demand and lead time variability.

Service Level: Percentage of customer demand fulfilled without stockouts, typically expressed as 90%, 95%, etc.

Standard Deviation (σ): Statistical measure of demand variability around the mean (13 units in simulation).

Stockout: Situation when demand cannot be fulfilled due to insufficient inventory.

Stockout Cost: Cost associated with not being able to fulfill customer demand (₹800 per unit in simulation).

Total Cost: Sum of all inventory-related costs including ordering, holding, stockout, and backorder costs.

Unit Cost: Price paid for each unit of inventory, subject to quantity discounts in the simulation.

Variability: Degree of fluctuation in demand or lead time, measured statistically.

Z-Score: Statistical value corresponding to a desired service level:

- 90% service level: $z = 1.28$
- 95% service level: $z = 1.65$
- 99% service level: $z = 2.33$
- 99.9% service level: $z = 3.09$

Key Inventory Management Formulas

	Concept	Formula	Explanation of Variables	Example
1	Economic Order Quantity (EOQ)	$EOQ = \sqrt{(2DS/H)}$	D = Annual demand (units) S = Ordering cost per order (₹) H = Annual holding cost per unit (₹)	D = 2,340, S = ₹600, H = ₹20 $EOQ = \sqrt{(2 \times 2340 \times 600 \div 20)} = \sqrt{1,404,000} = 374$ units
2	Reorder Point (ROP)	$ROP = \mu L + SS$ or $ROP = \mu L + z\sigma\sqrt{L}$	μ = Mean demand per period L = Lead time (periods) SS = Safety stock z = Z-score for service level σ = Std. deviation of demand	$\mu = 45, L = 3, z = 1.65, \sigma = 13$ $ROP = (45 \times 3) + (1.65 \times 13 \times \sqrt{3}) = 172$ units
3	Safety Stock (SS)	$SS = z \times \sigma \times \sqrt{L}$	z = Z-score σ = Std. deviation of demand L = Lead time	z = 1.65, $\sigma = 13, L = 3$ $SS = 1.65 \times 13 \times \sqrt{3} = 37$ units
4	Service Level	Service Level = $(\text{Total Demand Fulfilled} \div \text{Total Customer Demand}) \times 100$	—	Measures % of demand met without stockouts
5	Total Annual Inventory Cost	Total Cost = Ordering + Holding + Stockout + Backorder	—	Captures all relevant costs
6	Inventory Turnover	$COGS \div \text{Average Inventory Value}$ or $(\text{Annual Demand} \times \text{Unit Cost}) \div \text{Average Inventory Investment}$	—	Higher turnover = faster inventory movement
7	Average Inventory Level	Average Inventory = $(\text{Order Quantity} \div 2) + \text{Safety Stock}$	—	Cycle stock + safety stock
8	Cost Per Unit Served	Total Inventory Cost $\div$ Total Units Fulfilled	—	Indicates cost efficiency per unit delivered

